

SpaceCDF User Guide

AI-assisted Concurrent Design Facility — operator's reference

SpaceCDF · v6

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What SpaceCDF is

SpaceCDF is an AI-assisted Concurrent Design Facility for space mission design. It guides you from problem definition to a complete, verified CubeSat preliminary design with ECSS-compliant documentation. The tool sits at the centre of the CDF intensive described in the Course Plan and the Facilitator's Book.

The platform is implemented as a Python backend and a web frontend. You run it locally for solo work, or as a shared server for concurrent team sessions where each user occupies a CDF position (Power, AOCS, Comms, Payload, etc.).

EXPECTED READING

Expected reading Before opening the tool for the first time, skim the 3-Week Syllabus and Course Plan (this PDF set). They describe the workflow this guide assumes you know.

Lifecycle phases — what SpaceCDF helps you do where

Figure 3.1 — SpaceCDF carries you from Pre-Phase A (mission need) through end of Phase B (PDR). Operations (Phase E) is covered separately by the EOSAT-1 simulator and Week 3 packs.

Getting started

SYSTEM REQUIREMENTS

Component	Requirement
OS	macOS 13+, Ubuntu 22+, Windows 10/11 (WSL2 recommended)
Python	3.11 or newer
Node.js	18 LTS or newer
Browser	Chrome 120+ / Safari 17+ / Firefox 120+ — Edge 120+
RAM	8 GB minimum, 16 GB recommended
Disk	4 GB free for source + database

INSTALLATION (LOCAL)

```
git clone https://github.com/FNewland/SpaceCDF.git
cd SpaceCDF

# Backend
pip install -e packages/spacecdf-common
pip install -e packages/spacecdf-agents
pip install -e packages/spacecdf-kb
pip install -e packages/spacecdf-server
uvicorn spacecdf_server.app:app --reload --port 8000

# Frontend (separate terminal)
cd frontend && npm install && npm run dev
```

Open `http://localhost:5173` in your browser.

CONNECTING TO A REMOTE INSTANCE

Edit `frontend/vite.config.ts` and change the proxy target:

```
proxy: { '/api': 'http://REMOTE_HOST:8000' }
```

This routes API and WebSocket traffic to your shared CDF server. For Cyberrange-hosted teaching, the `REMOTE_HOST` is provided in Brightspace at start of Week 2.

Workflow at a glance

The five-step workflow follows the System-V (see [Course Plan](#)

Figure 2.2). Each step has a dedicated screen and a recognisable gate to the next:

Step	Screen	Gate to next	Course session
1 — Need	Stepper · Step 1 (Need)	Stakeholder map + objectives written	1.2
2 — Concept	Stepper · Step 2 (Concept)	Mission alternatives traded; archetype chosen	1.3 – 1.4
3 — Requirements	Stepper · Step 3 (Reqs)	Orbit / class chosen, design loop run	2.1 – 2.4
4 — Design	Tabbed workspace (5 groups)	Subsystems converged, interfaces clean	3.1 – 4.4
5 — Review	Gate Review tab	PDR exit criteria met	5.1 – 5.4

STEP 1 — MISSION NEED

1. Click **Step 1 (Need)** in the left stepper.
2. Enter your **problem statement**. WHAT not HOW — see Course Plan §2.1.
3. Add **stakeholders** with role and need.
4. Define **objectives** with measurable success criteria.
5. The tool auto-detects mission type from objective keywords.

STEP 2 — CONCEPT EXPLORATION

1. Click **Step 2 (Concept)**.
2. Enter trade parameters: GSD, revisit, coverage, latency, budget.
3. Click **Run Analysis** to evaluate space vs non-space alternatives.
4. Review scored results — the tool indicates whether space is justified.

STEP 3 — REQUIREMENTS

1. Click **Step 3 (Requirements)**.
2. **Orbit Advisor**: enter targets, click **Compute** → scored orbit candidates. Click **Use** to apply.
3. **Class Advisor**: enter known parameters (all optional). Click **Compute** → spacecraft class recommendation.
4. Set orbit, payload, and mission parameters in the form.
5. Click **Run Design (solo)** → 20 agents converge the design.

STEP 4 — DESIGN PHASE

After running the design, the tool switches to a tabbed design workspace organised into five groups:

Group	Tabs
Design	Dashboard · ConOps · Functions · Requirements · Interfaces
Analysis	Link Budget · Trade Studies · Optimizer · Cost
Verify	Compliance · V&V Matrix · Gate Review
Team	Positions · Q&A
Data	Exports · Parametric · Changes · Help

STEP 5 — REVIEW

Use the **Gate Review** tab to walk through PDR exit criteria. Resolve action items inline. Generate the ECSS document set from the **Exports** tab.

Key features

DASHBOARD

The main design view showing:

- **KPI cards:** mass margin, power margin, link margin, cost, sustainability, reliability, conflicts.
- **Mass waterfall** and **power profile** charts.
- **Budget breakdown:** per-subsystem mass / power with editable allocations.
- **Engineering budgets:** pointing (RSS), data (pipeline flow), timing (orbit timeline).
- **Spectrum selector:** choose licence type and frequency band.
- **Launch selector:** pick a provider with live mass-allocation update.
- **ECSS margin enforcement:** per-domain margin vs policy check.

A canonical mass budget rendered in the dashboard

Figure 3.2 — Mass distribution and the ECSS margin policy by phase. The dashboard's mass card shows this same data live.

EQUIPMENT BROWSER

Click **Equipment** in the header bar.

- **18 categories** grouped by domain (Power, AOCS, Comms, Propulsion, Structure, Data, Thermal, Integration).
- **Need annotations:** blue dot = required, circle = optional, dimmed = not needed.
- **Multiple selection:** select multiple items per category (e.g., 4 reaction wheels).
- **RF compatibility:** warning when selecting mismatched transponder / antenna bands.
- **Live budget bar:** running mass / power / cost totals.
- **Compare mode:** side-by-side comparison of up to 3 components.

TRADE STUDIES

Two modes:

- **Parametric Sweep** — vary one parameter, see effect on key metrics (chart).
- **Tabular Trade** — define criteria with weights, add options, score quantitatively or qualitatively (low / medium / high), see ranked results.

LINK BUDGET TOOL

Full interactive cascade — TX power → antenna gain → losses → free-space path loss → atmospheric / pointing / polarisation → RX gain → G/T → C/N₀ → Eb/N₀ → margin. Each term editable, margin colour-coded.

Link-budget waterfall — S-band downlink, 1500 km slant

Figure 3.3 — A canonical link-budget waterfall as rendered by SpaceCDF. Required C/N₀ and link margin sit at the bottom-right.

OPTIMIZER

- **Single objective:** minimise mass, cost, or maximise link margin.
- **Pareto (NSGA-II):** multi-objective with up to 10 objectives.
- **Mission-type aware:** variables auto-filtered (no propulsion vars without propulsion).
- **Sensitivity (Morris):** ranks which parameters most influence the objective.

EXPORTS

Generates 18+ document types:

- **ECSS:** MRD, TS, IRD, SEMP, RMP, ConOps, VP, Test Plan.
- **Regulatory:** ITU API, IARU coordination, RSSSA, export control, COPUOS, EOL report.
- **Design data:** parametric model data, duty cycles, consistency check, margin enforcement.

CROSS-TOOL REACTIVITY

- **Design state bar:** shows when design is outdated after any parameter change.
- **Auto-reconverge:** optional toggle — re-runs design automatically after each edit.
- **Conflict review:** modal appears when critical conflicts detected; requires resolution.
- **Impact preview:** shows which agents and budgets a change will affect before accepting.
- **Change audit:** full history with undo capability.

PROGRESSIVE LEVEL UNLOCKING

The tool follows the System-V (see Figure 3.4) with 5 progressive levels:

- **Level 0** — Help only (define mission need first).
- **Level 1** — Mission architecture (after need defined). ConOps, Functions, Requirements.
- **Level 2** — System architecture (after design run). Architecture options, Interfaces, Budgets, Trade Studies, Project Management.
- **Level 3** — Subsystem design (after architecture selected). Link Budget, Optimizer, Cost, Equipment Browser.
- **Level 4** — Verification (after subsystem design). Compliance, V&V Matrix, Gate Review.

The level indicator bar shows your current progress and what to do next.

System-V — what SpaceCDF unlocks at each level

Figure 3.4 — System-V model. Level unlocking in SpaceCDF tracks the left-hand decomposition leg.

INTERACTIVE MISSION ARCHITECTURE EDITOR

In the **ConOps** tab, a drag-and-drop diagram editor lets you build your mission architecture:

- **6 node types** — Satellite · Ground Station · Processing · User · Sensor · GNSS/External.

- Drag nodes to position, connect with labelled lines.
- Architecture drives what systems need to be defined at the next level.

SYSTEM ARCHITECTURE SELECTION

In the **Architecture** tab:

- **Top half** — select architecture options for each of 8 subsystems (EPS, AOCS, TT&C, Thermal, Structure, Propulsion, OBC, Ground).
- **Bottom half** — auto-generated block diagram showing subsystems and their connections.
- Each selection derives requirements (tagged performance / interface / budget / functional).

ENGINEERING BUDGETS (UNIFIED VIEW)

The **Eng. Budgets** tab shows all 8 budgets in one place:

Budget	Source	Margin (Phase A / B / C)
Mass	Equipment + structure model	20 % / 15 % / 10 %
Power	Equipment + duty-cycle model	25 % / 20 % / 10 %
Link	Link-budget cascade	6 dB / 3 dB / 3 dB
Pointing	RSS pointing tree	30 % / 20 % / 10 %
Δv	Propulsion model	25 % / 15 % / 10 %
Volume	Cal Poly CDS envelope	10 % / 5 % / 2 %
Data	Pipeline flow model	30 % / 20 % / 10 %
Cost	Parametric CER + COTS	30 % / 20 % / 15 %

Per-subsystem breakdown shows margin source (COTS vs new design) and rolls up from equipment selections.

CONSTRAINT PROPAGATION ENGINE

187 design-point interconnections detected automatically:

- When any parameter changes, the engine identifies ALL affected budgets.
- Shows resolution options for violations with cross-budget trade-off analysis.
- Detects circular dependencies (trade-off loops requiring team decision).

PROJECT MANAGEMENT

The **Project Mgmt** tab provides:

- **5x5 risk matrix** — interactive, colour-coded by L × C score. See Course Plan Figure 2.11.

- **Schedule** — 10 milestones from MCR through commissioning.
- **WBS** — 9 work packages with effort hours and status tracking.
- **Project Manager** position included.

WORD DOCUMENT EXPORT

ECSS documents can be exported as editable `.docx` files:

- Click **Word** button next to MRD, ConOps, or VP in the **Exports** tab.
- Documents populated from live design state.
- Editable in Microsoft Word or LibreOffice.

SESSION PERSISTENCE

Design state persists across page refreshes:

- All data saved to browser `localStorage` automatically.
- No need to “save” — it’s always saved.
- To clear: browser DevTools → Application → Local Storage → delete `spacecdf-design-state`.

Concurrent design sessions

1. Click **Start Session** in the header.
2. Select your position(s) and enter a display name.
3. Share the URL with team members — they join the same session.
4. Edits propagate in real-time via WebSocket.
5. Each position can only edit their owned parameters.
6. Convergence runs automatically after each edit.

Editing parametric data

The tool’s sizing models use CubeSat-calibrated data. To view and edit:

1. Go to **Parametric** tab.
2. Four sub-tabs: Mass Fractions · Cost Fractions · Power Duty Cycles · SA Power.
3. Values are per spacecraft class (nano, micro, small, medium, large).
4. Sources cited alongside each table.

To edit the source data directly: `packages/spacecdf-common/src/spacecdf_common/physics/heritage_mass.py`.

Updating equipment database

Component data is in YAML files: `packages/spacecdf-kb/src/spacecdf_kb/data/components/`.

To add a component:

```
- id: unique-id
  name: "Component Name"
  manufacturer: "Vendor"
  mass_kg: 0.15
  power_w: 8.0
  cost_keur: 30
  trl: 8
  frequency_band: S # for RF components
```

The API also supports import: `POST /api/lifecycle/equipment/import`.

Tab-by-tab feature inventory

Tab	Group	Purpose
Dashboard	Design	Live KPI overview
ConOps	Design	Mission architecture editor & timeline
Functions	Design	Function tree, allocation matrix
Requirements	Design	SMART requirements, parent/child
Interfaces	Design	N ² matrix, interface conflict resolver
Architecture	Design	Architecture selection, derived reqs
Eng. Budgets	Design	All 8 budgets in one place
Link Budget	Analysis	Interactive cascade, margin colouring
Trade Studies	Analysis	Parametric sweep + tabular
Optimizer	Analysis	NSGA-II, Morris sensitivity
Cost	Analysis	CER + WBS roll-up
Compliance	Verify	Requirement compliance matrix
V&V Matrix	Verify	Verification method (A/T/R/I) per req
Gate Review	Verify	PDR exit criteria, action items
Positions	Team	Position assignments + responsibilities
Q&A	Team	Position-specific design questions
Project Mgmt	Team	Risk · schedule · WBS
Exports	Data	ECSS, regulatory, design-data exports
Parametric	Data	Tunable sizing-model coefficients
Changes	Data	Audit log + undo
Help	Data	Inline help index

Status-bar legend

The bottom status bar uses 4 colour bands and 6 icons:

Colour	Meaning	Action
Green	All margins met, no conflicts	Continue
Amber	Margin within 10 % of policy	Tighten or accept and document
Red	Margin violated or critical conflict	Resolve before next gate
Grey	Design out of date — re-run	Click “Run Design”

Icons:  active design ·  nominal ·  warning ·  critical ·  reconverging ·  export ready.

File-format support matrix

Format	Import	Export	Notes
YAML (mission, equipment)	✓	✓	Canonical interchange
JSON (parameters)	✓	✓	API native
CSV (BOM, requirements)	✓	✓	Excel-friendly
Word (.docx)	—	✓	ECSS document set
PDF	—	✓	Generated from .docx
Markdown	—	✓	Read-only design pack
TLE	✓	✓	Orbit interchange
KML	—	✓	Ground tracks for visual review

Keyboard shortcuts

Shortcut	Action
Ctrl + Enter	Run design
Ctrl + S	Force save (no-op — always saved)
Esc	Close modal
⌘/Ctrl + K	Quick-search any tab
F2	Focus dashboard
F3	Focus equipment browser
F4	Focus optimiser
⌘/Ctrl + Z	Undo (uses Change Audit)
⌘/Ctrl + Shift + Z	Redo

Troubleshooting

Issue	Cause	Solution
“Socket not connected”	No active session	Start or join a session first
Design shows no results	Design not run	Click Run Design in Step 3
Requirements tab empty	No study ID	Run design at least once
Optimizer won’t start	No active session	Start a session, or use solo mode
Equipment browser greyed out	Frontend ↔ backend out of sync	Reload the page; check CORS
ECSS export fails	Design has open conflicts	Resolve in Conflict Review modal
Word export looks wrong	LibreOffice quirks	Open in Word; missing fonts: install Work Sans/Spectral
TLE import rejected	Wrong epoch / line length	Use the canonical 2-line format

Error-message reference

Code	Message	What it means	What to do
E_NEED_001	“Problem statement required”	Step 1 incomplete	Write the WHAT in Step 1
E_REQ_017	“SMART check failed: vague verb”	Requirement not testable	Use shall + measurable verb
E_INT_004	“Interface conflict — voltage mismatch”	Two subsystems disagree on bus voltage	Decide bus voltage at architecture level
E_BUD_022	“Mass margin violated”	Total > policy	Trade structure / payload / propellant
E_OPT_011	“Optimizer not converging”	Search space empty	Widen variable bounds
E_EXP_003	“ECSS export blocked: open conflicts”	Conflicts not resolved	Resolve in Conflict Review

Glossary

Term	Meaning
AOCS	Attitude & Orbit Control System
AOS	Acquisition of Signal (start of pass)
BOM	Bill of Materials
CDF	Concurrent Design Facility
CER	Cost Estimating Relationship
ConOps	Concept of Operations
CDS	(Cal Poly) CubeSat Design Specification
DoD	Depth of Discharge (battery)
ECSS	European Cooperation for Space Standardization
EIRP	Effective Isotropic Radiated Power
EPS	Electrical Power Subsystem
FDIR	Fault Detection, Isolation, and Recovery
FOV	Field of View
FSPL	Free-Space Path Loss
GSD	Ground Sample Distance
ICD	Interface Control Document
IRD	Interface Requirements Document
ITU	International Telecommunication Union
LEOP	Launch and Early Orbit Phase
LOS	Loss of Signal (end of pass)
MCR	Mission Concept Review
MoE / MoP	Measure of Effectiveness / Performance
MRD	Mission Requirements Document
OBDAH	On-Board Data Handling
OBC	On-Board Computer
PCDU	Power Conditioning & Distribution Unit
PDR	Preliminary Design Review
PUS	(CCSDS) Packet Utilization Standard
RAAN	Right Ascension of Ascending Node
RACI	Responsible / Accountable / Consulted / Informed
RSSSA	Remote Sensing Space Systems Act (Canada)

Term	Meaning
RW	Reaction Wheel
SAR	Synthetic-Aperture Radar
SCM	Space Mission Engineering (textbook)
SMAD	Space Mission Analysis & Design (textbook)
SMART	Specific · Measurable · Achievable · Relevant · Traceable
SSO	Sun-Synchronous Orbit
TC	Telecommand
TCA	Time of Closest Approach
TM	Telemetry
TPM	Technical Performance Measure
TRL	Technology Readiness Level
TT&C	Tracking, Telemetry & Command
WBS	Work Breakdown Structure

Reference list

- **NASA SEH** — <https://www.nasa.gov/reference/systems-engineering-handbook/>
- **NASA CubeSat 101** — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_c-sli_cubesat_101_508.pdf
- **ECSS standards** — <https://ecss.nl/>
- **Cal Poly CDS Rev 14** — <https://www.cubesat.org/cds-announcement>
- **CCSDS PUS** — <https://public.ccsds.org/Pubs/660x0g3.pdf>
- **ITU Radio Regulations** — <https://www.itu.int/pub/R-REG-RR>
- **ISED CPC-2-6-02** — <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en>
- **SpaceCDF source** — <https://github.com/FNewland/SpaceCDF>